

STAT 24620 / FINM 34700 / STAT 32950
Homework 2

Due: 04/19/2026 11:59pm

Problem 1: hierarchical clustering with different distance choices

The vocabulary richness of an ancient text (Book of Lamentations) can be quantitatively described by counting the words used once, the words used twice, and so forth. Based on these counts, a linguist proposed the following distances between Chapters 1–5:

$$D = \begin{pmatrix} 0 & 0.76 & 2.97 & 4.88 & 3.86 \\ 0.76 & 0 & 0.80 & 4.17 & 1.96 \\ 2.97 & 0.80 & 0 & 0.21 & 1.51 \\ 4.88 & 4.17 & 0.21 & 0 & 0.51 \\ 3.86 & 1.96 & 1.51 & 0.51 & 0 \end{pmatrix}.$$

- (a) Perform agglomerative hierarchical clustering using single linkage, complete linkage, and average linkage.
- (b) Compare the three dendrograms, including whether merge heights are on comparable vertical scales.
- (c) Cut each tree at $K = 3$ clusters and report the three clusters from each linkage method.
- (d) Briefly compare and comment: what is your main takeaway about linkage choice?

Problem 2: clustering on women's track records

Use the dataset `ladyrun25.dat` provided in HW1, containing national women's running records:

Country, 100m, 200m, 400m, 800m, 1500m, 3000m, Marathon.

Treat the running-time variables as the clustering features. You may use either the raw times or standardized times, but state your choice clearly and use it consistently throughout.

- (a) Compute all pairwise Euclidean distances between countries (do not print the full matrix). Give the two countries (translate into true names) with the maximum distance and the two countries with the minimum distance, along with the distance values.
- (b) To illustrate the mechanics of K -means, consider $K = 2$ and one Lloyd iteration. Choose two countries as initial centroids, and perform *one full Lloyd iteration*:

- (i) assignment step,
- (ii) centroid update step.

Report the updated centroids after this iteration. Based on the updated centroids, do you expect the assignments to change in the next iteration? Briefly justify.

- (c) For the same setup as in part (c), compute the within-cluster sum of squares

$$\text{WCSS} = \sum_{i=1}^K \sum_{x \in C_i} \|x - m_i\|_2^2$$

before and after this Lloyd iteration. Verify that the objective does not increase, and explain briefly why this should be true in general.

- (d) Run K -means clustering for $K = 2, \dots, 8$. Plot the average silhouette width versus K , and identify a reasonable choice of K . Briefly justify your answer.
- (e) Fit Gaussian mixture models, allowing both the number of components K and the covariance structure to be selected by BIC. Report the selected number of clusters K , the selected covariance model, the cluster sizes, and the countries in the three smallest clusters. Briefly interpret the selected covariance structure (e.g., spherical, diagonal, or full; equal vs. varying volume, shape, and orientation).
- (f) Compare the clustering results from K -means and the Gaussian mixture model. In a short paragraph, discuss whether the two methods suggest similar numbers of clusters, identify any countries that are clustered differently, and explain why the two methods may produce different results on this dataset. Your answer should refer to the underlying assumptions of the two methods.

Problem 3: GMM parameter counting and constrained EM updates

Assume a K -component Gaussian mixture in $\mathbb{R}^p$:

$$p(x) = \sum_{k=1}^K \pi_k \phi_p(x \mid \mu_k, \Sigma_k), \quad \sum_{k=1}^K \pi_k = 1, \quad \pi_k > 0.$$

- (a) For each covariance specification below, derive the total number of free parameters:

- (i) full covariance by component (Σ_k unrestricted and distinct),
- (ii) shared full covariance ($\Sigma_k = \Sigma$ for all k),
- (iii) diagonal covariance by component,
- (iv) spherical covariance by component ($\Sigma_k = \sigma_k^2 I_p$).

- (b) Now, assume a **shared full covariance** GMM:

$$\Sigma_1 = \dots = \Sigma_K = \Sigma.$$

Let $\Theta^{(t)} = \{\pi_k^{(t)}, \mu_k^{(t)}, \Sigma^{(t)}\}_{k=1}^K$ denote the current parameter estimates. Write the E-step responsibilities

$$\gamma_{ik} = \Pr(Z_i = k \mid x_i, \Theta^{(t)}).$$

- (c) Assume the same model as in (b), derive M-step updates for $\pi_k^{(t+1)}$, $\mu_k^{(t+1)}$, and $\Sigma^{(t+1)}$.
- (d) In 3–5 sentences, compare the statistical/computational tradeoff between “full covariance by component” that we taught in class and “shared full covariance” here.

Problem 4: loadings in PCA and CCA

In the lecture of PCA, we referred to the entries of the principal component direction $\mathbf{v} = (v_1, \dots, v_p)$ as *loadings*. More generally, loadings are defined as correlations between observed variables and the components.

- (a) (PCA): Let $X \in \mathbb{R}^p$ be centered and standardized random vector, and let

$$Z = \mathbf{v}^\top X$$

be the first principal component with eigenvalue λ .

Show that

$$\text{Corr}(X_j, Z) = \sqrt{\lambda} v_j.$$

Conclude that, in PCA, loadings (as correlations) are proportional to the weights.

- (b) (CCA): Let $U = \mathbf{a}^\top X$ be a canonical variate.

Show that

$$\text{Cov}(X, U) = \Sigma_{XX} \mathbf{a}.$$

Express the loading vector

$$(\text{Corr}(X_1, U), \dots, \text{Corr}(X_p, U))^\top$$

in matrix form (in terms of Σ_{XX} and $\mathbf{a}$), and deduce that, in general, the loadings $\text{Corr}(X_j, U)$ are not proportional to the weights a_j .

- (c) Consider the case $p = 2$. Give a concrete example (specifying Σ_{XX} and $\mathbf{a}$ qualitatively or numerically) where a variable can have a small weight a_j but a relatively large loading.

Problem 5: CCA on stiffness data

Use the dataset `stiffness.DAT`, with four different measures of stiffness x_1, x_2, x_3, x_4 measured on $n = 30$ boards. The first measurement involves sending a shock wave down the board, the second measurement is determined while vibrating the board, and the last two measurements are obtained from static tests. The squared distances are also included as the last column in the data.

Define

$$X = (x_1, x_2)^\top \quad (\text{dynamic measures}), \quad Y = (x_3, x_4)^\top \quad (\text{static measures}).$$

- (a) Perform CCA on the full dataset and report the sample canonical correlations $(\hat{\rho}_1, \hat{\rho}_2)$.

- (b) Write the first pair of canonical variates, U_1 and V_1 , as linear combinations of the variables in X and Y .
- (c) Produce two scatterplots of canonical scores: (U_1, V_1) and (U_2, V_2) . Briefly comment on the strength and pattern of association shown in each plot.
- (d) Report the pair-1 loadings on both sides:

$$\text{Corr}(x_1, U_1), \text{Corr}(x_2, U_1), \text{Corr}(x_3, V_1), \text{Corr}(x_4, V_1).$$

Use loadings, rather than raw weights alone, for interpretation.

- (e) Repeat the following procedure 20 times: randomly split the data into 70% training and 30% test sets, fit CCA on the training set only, and compute the test-set correlations for the training-derived score pairs (U_1, V_1) and (U_2, V_2) .
- (f) Summarize the 20 training-set canonical correlations and the 20 corresponding test-set correlations for each pair (for example, by reporting the mean and standard deviation, or by using a boxplot/table). Compare the training and test results, and briefly comment on whether the observed relationships appear to generalize or may reflect overfitting.